

COE3001 Mechanics of Deformable Bodies - Final Review.

1. Roadmap

Basic deformations { Tension/compression.
Torsion
Bending

Combined load : { Stress/strain transformation.
Generalized Hooke's law

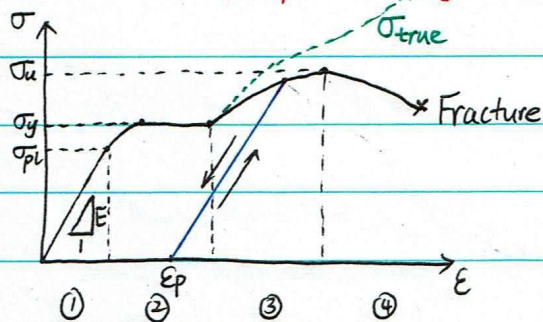
Stability : Column buckling

load $\longleftrightarrow$ reaction
Force/Moment $\longleftrightarrow$ Displacement/Rotation

stress $\longleftrightarrow$ Strain.

Hooke's Law : $\sigma = E \epsilon$
(linear elasticity) $\tau = G \gamma$
 $G = \frac{E}{2(1+\nu)}$
Young's modulus
Poisson's ratio

2. Mechanical properties of materials.



- ① linear region
- ② perfect plasticity
- ③ strain hardening
- ④ Necking

$\sigma_{true} = \sigma(1 + \epsilon)$
 $\epsilon_{true} = \ln(1 + \epsilon)$
true value : change A

3. Basic deformations.

3.1 Tension / compression.

Force (F) $\longleftrightarrow$ displacement (u)
 $\sigma = \frac{F}{A}$ $\epsilon = \frac{\Delta L}{L}$
 $\sigma(x) = \frac{F(x)}{A(x)}$ $\epsilon = \frac{du}{dx}$
(normal) stress (σ) $\longleftrightarrow$ strain (ϵ)
 $\sigma(x) = E \epsilon(x)$ $\sigma = E \epsilon$

$$\Rightarrow \delta = \Delta L = \frac{FL}{EA}$$

$$u(x) = \int_0^x \frac{F(x)}{EA(x)} dx$$

$$\delta = u(L) = \int_0^L \frac{F(x)}{EA(x)} dx$$

3.2 Torsion

Torque (T) $\longleftrightarrow$ Angle of rotation (ϕ)
 $\tau = \frac{Tp}{I_p}$ $\gamma = \frac{d\phi}{dx}$
 $\tau(x) = \frac{T(x)p}{I_p(x)}$ $\gamma = \frac{d\phi}{dx}$
(Shear) stress $\longleftrightarrow$ strain
 $\tau(x) = G \gamma(x)$ $\tau = G \gamma$

$$\Rightarrow \phi = \frac{TL}{G I_p}$$

$$\phi(x) = \int_0^x \frac{T(x)}{G I_p(x)} dx$$

$$\phi = \phi(L) = \int_0^L \frac{T(x)}{G I_p(x)} dx$$

3.3 Bending

Bending moment (M)

deflection (v)

$$\sigma = -\frac{My}{I_z}$$

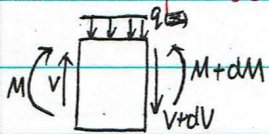
$$\epsilon = -\frac{y}{\rho} = -yK$$

(normal) stress $\longleftrightarrow$ strain $\sigma = E\epsilon$

$$\Rightarrow EI K = M \iff EI v'' = M$$

Differential equation of deflection.

3.3.1 Relationships between $q(x)$, $V(x)$, and $M(x)$



$$\frac{dV}{dx} = q(x), \quad q(x) \text{ downward}$$

$$\frac{dM}{dx} = V(x)$$

$$EI \frac{d^2 v}{dx^2} = M$$

$$EI \frac{d^3 v}{dx^3} = \frac{dM}{dx} = V$$

$$EI \frac{d^4 v}{dx^4} = \frac{dV}{dx} = -q$$

3.3.2 Bending Shear stress in beams.

$$\tau = \frac{V}{Ib} \int y dA = \frac{V}{2I} \left(\frac{h^2}{4} - y^2 \right) \text{ for rectangular beam}$$

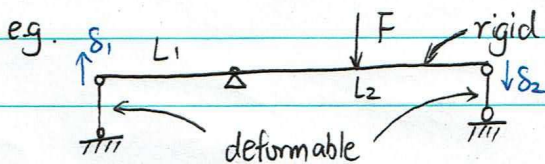
$$\Rightarrow \tau_{\max} = \frac{3}{2} \tau_{\text{avg}} = \frac{3}{2} \frac{V}{A}$$

3.4 Statically indeterminate problems.

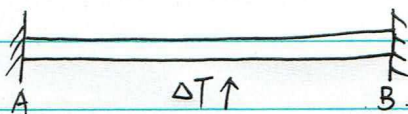
* degree of static indeterminacy = # of overconstraints. (unknown force).

= # of unknowns - # of equilibrium equations.

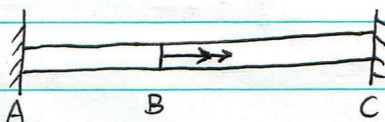
* overconstraints give compatibility equations. (usually in terms of displacements)



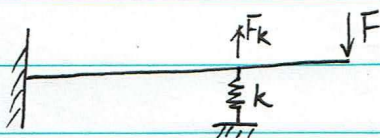
$$\delta_1 / L_1 = \delta_2 / L_2$$



$$\delta_{AB} = 0 = \delta_T + \delta_F$$



$$\phi_{AB} = -\phi_{BC} \quad \text{or} \quad \phi_{AC} = 0$$



$$F_k = k \delta_k \quad (\text{sometimes in terms of unknown force directly})$$

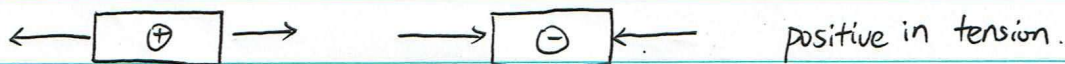
3.4.1 Standard procedure for solving statically indeterminate problems.

1. FBD
2. equilibrium equations
3. calculate degree of static indeterminacy.
4. list all compatibility equations.
5. convert compatibility equation into equation of unknowns.
6. solve equilibrium equation & compatibility equation simultaneously. $\Rightarrow$ reaction forces.
7. calculate stress / strain / displacement / stability etc.

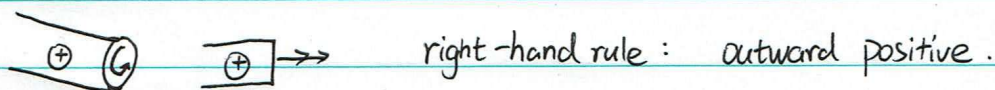
4. Methods to get internal loads.

4.1 Sign convention for internal loads.

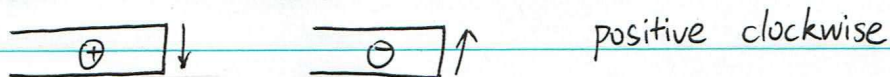
Normal force :



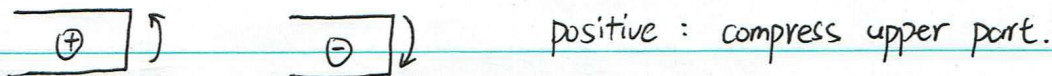
Torsional moment :



Bending shear force

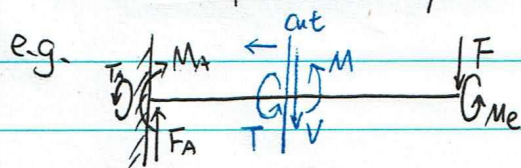


Bending moment



4.2 Method of sections

1. make a cut
2. assume internal loads in its positive direction.
3. solve equilibrium equation to get value of internal load.



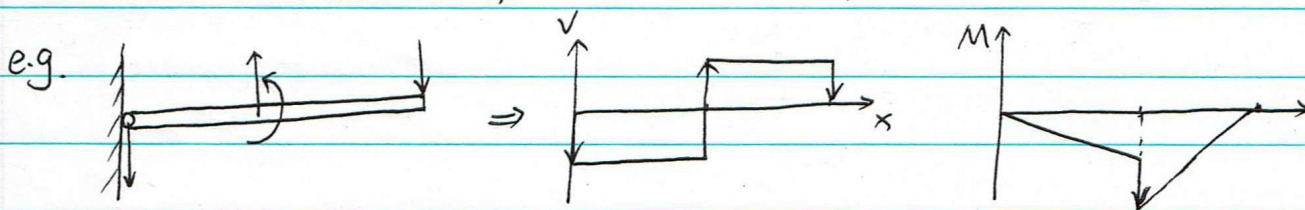
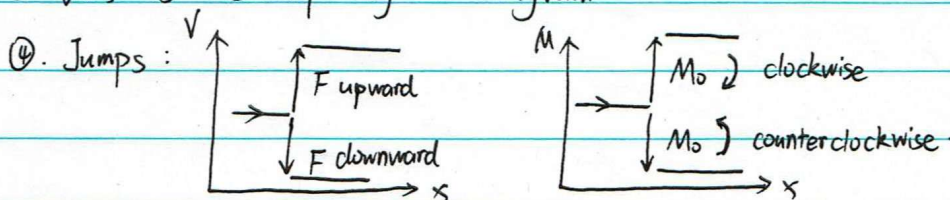
$$T = T_A$$

$$M = M_A$$

$$V = F_A$$

4.3 Method of Jumps.

- ① For beams, only from left to right
- ② $-q(x)$ is the slope of V-diagram. ($q(x)$ downward)
- ③ $V(x)$ is the slope of M-diagram
- ④ Jumps:



5. Methods for solving beam deflections.

5.1 Method of successive integrations.

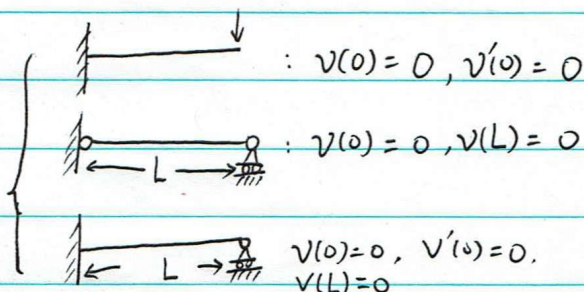
after solving reaction forces, get internal bending moment,

$$① \quad EI v'' = M$$

$$② \quad EI v' = \int_0^x M(x) dx + C_1$$

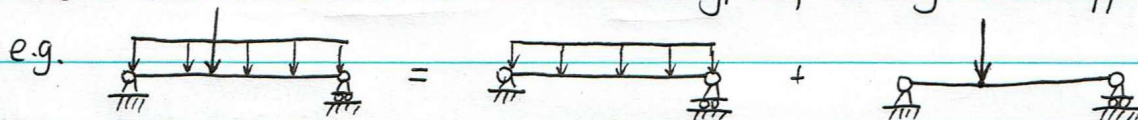
$$③ \quad EI v = \int_0^x \left[\int_0^x M(x) dx \right] dx + C_1 x + C_2$$

④ apply boundary conditions to solve C_1 & C_2



5.2 Method of superposition

Break the loads into several common types of loadings and supports.



6. Stress / strain transformation.

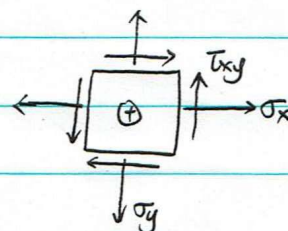
$$\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

Mohr's circle:

$$(\sigma - \sigma_{avg})^2 + \tau^2 = R^2, \quad \sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}, \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

* τ positive downward.



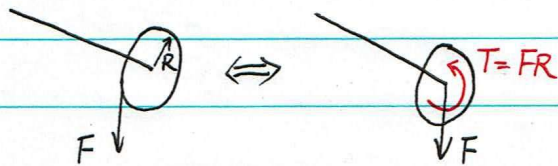
for strain:

$$\sigma \rightarrow \epsilon$$

$$\tau \rightarrow \frac{\gamma}{2}$$

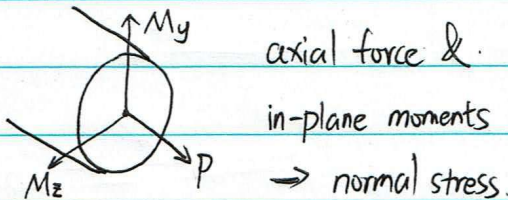
6.1 Combined load & strength theories.

move the applied load:

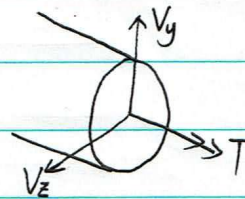


replace the eccentric load
by concentric load + equivalent moment.

Internal loads:



axial force &
in-plane moments
→ normal stress.



shear force &
axial moment
⇒ shear stress.

- Maximum tensile stress theory: $\sigma_{r1} = \sigma_1 \leq [\sigma]$
- Maximum tensile strain theory: $\sigma_{r2} = \sigma_1 - \nu(\sigma_2 + \sigma_3) \leq [\sigma]$
- Maximum shear stress theory: $\sigma_{r3} = \sigma_1 - \sigma_3 \leq [\sigma]$
- * Maximum distortional strain energy density theory: $\sigma_{von} = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_3)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_1 - \sigma_2)^2]} \leq [\sigma]$

7. Columns.

$$P_{cr} = \frac{\pi^2 EI}{(Le)^2} \quad Le = KL$$

- long columns ($\lambda > \lambda_p$)

$$\sigma_{cr} = \frac{\pi^2 E}{\lambda^2}, \quad \lambda = Le/r, \quad r = \sqrt{\frac{I_z}{A}}$$

- intermediate columns: ($\lambda_s < \lambda < \lambda_p$).

$$\lambda_p = \sqrt{\frac{\pi^2 E}{\sigma_{pl}}}$$

$$\lambda_s = \frac{a - \sigma_s}{b}$$

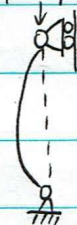
if $\sigma_{cr} = a - b\lambda$. (linear form).

- Short columns. ($\lambda < \lambda_s$):

no stability issues.

* Safety factor for stability: $\frac{F}{A} \leq \frac{\sigma_{cr}}{n_{st}} = [\sigma]_{st}$.

• pin-pin



$K=1$

• fixed-free



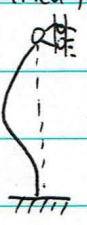
$K=2$

• fixed-fixed



$K=0.5$

• fixed-pin



$K=0.7$